

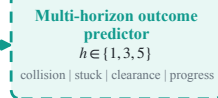
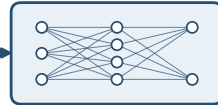
Local observation

Modality encoders

Shared representation

Predictive actor-critic

Command



Bounded 3-D velocity

$$\mathbf{a}_t^G = v_{\text{lim}}(2\tilde{\mathbf{a}}_t - 1)$$

$$v_{\text{lim}} = 2 \text{ m/s}$$

Training-only supervision

Joint optimization

$$\mathcal{L} = \mathcal{L}_{\text{PPO}} + \lambda_{\text{MH}} \mathcal{L}_{\text{MH}}$$

shared predictive representation



Parallel curriculum rollouts

dynamic + non-convex obstacles



joint representation update

current $\tilde{\mathbf{a}}_t$

action